Ref. Ares(2024)2369551 - 31/03/2024

Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

Project funded by the European Innovation Council and SMEs Executive Agency

D3.3 Methods to predict and quantify uncertain scenario

Work Package: WP3

Authors: Kevin Tom Kurian, Mohsen Alirezaei

Project Title: Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

Project Acronym: SymAware

Grant Agreement number: 101070802

Project Start Date: 1st October 2022

Duration: 36 months

Project Coordinator: Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

Siemens Industry Software Netherlands BV (SISW)

Document Identifier: D3.3

Issue: 1.0

Issue Date: 30.03.2024

Prepared by: TUE **Approved:** MPI-SWS

Dissemination Status: Public

History Chart:

Issue	Date (DD/MM/YYYY)	Updated Sections	Cause of Change	Implemented by
0.1	15.02.2024	All sections	New Document	Kevin Kurian, Mohsen Alirezaei (TUE)
0.2	01.03.2024	All sections	Review	Erjen Lefeber, Mohsen Alirezaei (TUE)
1.0	30.03.2024	All sections	Final Document	Mohsen Alirezaei (TUE)

All rights reserved

The document is proprietary of the SymAware consortium members listed on the front page of this document. The document is supplied on the express understanding that it is to be treated as Unrestricted and may not be used or disclosed to others in whole or in part for any purpose except as expressly authorized in terms of Grant Agreement number 101070802. The SymAware consortium makes no warranty for the information contained in this document; neither does it assume any legal liability or responsibility for the accuracy, completeness or usefulness of this information. Company or product names mentioned in this document may be trademarks or registered trademarks of their respective companies.

Contents

List of Figures 4
List of Abbreviations 5
1 Introduction 6
1.1 SymAware: High-level description of the project 6
1.2 Purpose and contents of this report 8
2 Scenario definitions and behavioral modeling 9
2.1 Background 9
2.2 Literature review on scenario definition 9
2.3 Behavior Modeling Approach 11
2.4 Scenario Terminology Definition 12
2.5 Modeling a system of interest 13
3 Dissimilarity metric 15
3.1 Literature review 15
3.2 Metric Foundation 17
3.2.1 Concept of Dissimilarity 17
3.2.2 Abstraction with a Mesh grid 17
3.2.3 Dissimilarity at Critical Scene 19
3.2.4 Assumptions for the dissimilarity metric 19
3.2.5 Properties of the dissimilarity metric 19
3.3 Dissimilarity metric formulation 21
3.3.1 Critical Scene Calculation 21
3.3.2 Discrete and Continuous Variables 22
3.3.3 Dissimilarity Calculation 24
4 Dissimilarity metric implementation 25
4.1 Dissimilarity for the modeled SOI 25
4.2 Clustering scenarios using dissimilarity 26

5 Conclusions and follow-up 29

Bibliography 31

List of Figures

Figure 1 The novel cogitative architecture developed in SymAware 7

Figure 2 The main (technical) work packages of SymAware 7

Figure 3 Modeled System of Interest (SOI) Σ 14

Figure 4 A sample representation of meshing 18

Figure 5 Representation of Relative heading θrel and angle of minimum distance in the local frame of reference ϕc. Counter-clockwise measurement is considered positive, and clockwise measurement is considered negative. 23

Figure 6 Flowchart for the dissimilarity metric proposed, with averaging by addition for the continuous variables 24

Figure 7 Base scenario trajectories for all actors in the modeled SOI 25

Figure 8 Collision modes for scenario path in Figure 7 26

Figure 9 Minimum distance and continuous variables (θ_{rel}, ϕ_c) as a function of velocity ratio of actors 27

Figure 10 Minimum distance and continuous variables (θ_{rel}, ϕ_c) as a function of velocity ratio of actors, with clustering 28

List of Abbreviations

MAS	Multi-Agent Systems
SOTIF	Safety Of The Intended Functionality
ADS	Automated Driving Systems
SBT	Scenario-Based Testing
ODD	Operational Design Domain
AV	Automated Vehicle
SOI	System of Interest
VOI	Variables of Interest

1 Introduction

1.1 SymAware: High-level description of the project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi- agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and humans. The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks. The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterize and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through

1. logical characterization of awareness using symbolic methods,
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the “Awareness Inside” Challenge of the European Innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterization and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

Figure 1 shows a schematic view of the SymAware new cognitive awareness architecture. Together the objectives of this proposal will achieve the depicted situational awareness for multi-agent settings in which the situational awareness is related to the different awareness dimensions and to contemporary methods for probabilistic modelling, perception, estimation, and (data-driven) computation.

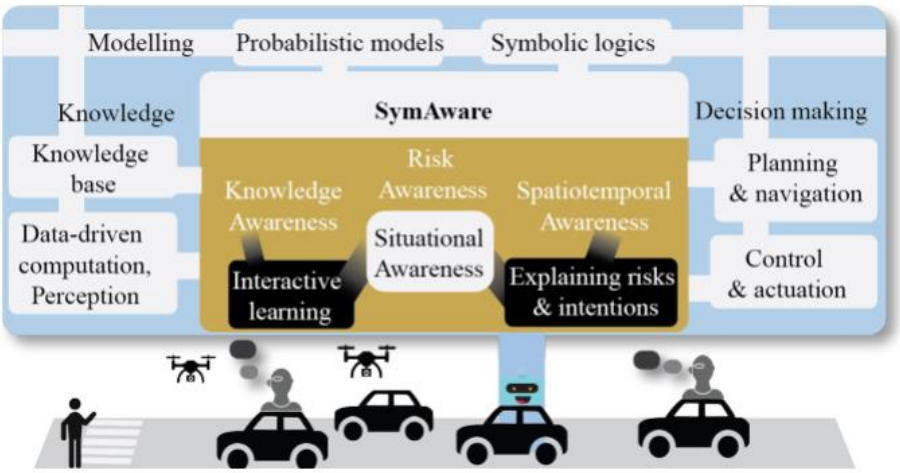


Figure 1 The novel cogitative architecture developed in SymAware

Figure 2 shows the main technical work packages of SymAware. The comprehensive framework of situational awareness and its logical characterization (Objective 1) will be developed in WP1. The spatiotemporal awareness of Objective 2 will be studied in WP2. Risk and knowledge awareness (Objectives 3 and 4) will be studied in WP3 and WP4, respectively. The automotive and aviation use cases (Objective 5) will be developed in WP5. Finally, the human-robot interaction and the ethical requirements (Objective 6) will be studied in WP6.

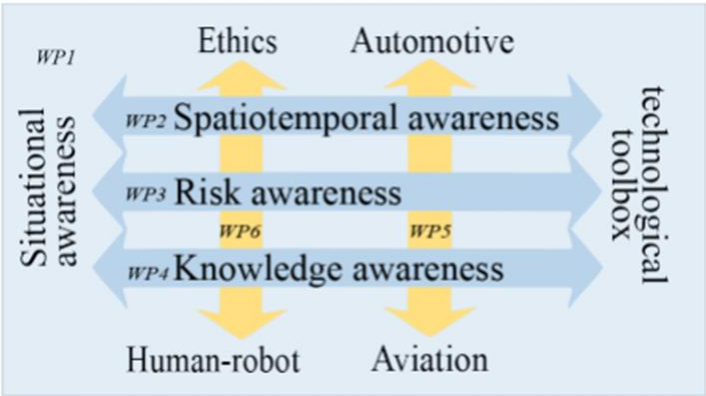


Figure 2 The main (technical) work packages of SymAware

1.2 Purpose and contents of this report

Deliverable D3.3 from WP3 concerning risk awareness focuses on two primary areas: 1) the mathematical modeling of scenarios and 2) the establishment of a metric for assessing scenario dissimilarity. Mathematical modeling enables the abstraction of scenarios into modular forms, formalizing them mathematically to encompass all potential environmental situations. Meanwhile, the dissimilarity metric offers a mathematical tool for comparing scenarios. These formalized scenarios and the dissimilarity metric lay the groundwork for the optimization process in WP5, aimed at identifying unknown unsafe scenarios in accordance with SOTIF (Safety Of The Intended Functionality) standards, crucial for assessing the safety of automated driving systems (ADS).

While various definitions exist in the literature to standardize scenario-related terminology, these semantic definitions can lead to ambiguous interpretations. To resolve this ambiguity, precise terminology formulation in mathematical language is necessary. However, a universally accepted mathematical definition for scenarios and related terminology that is both consistent and rigorous is lacking. This deliverable addresses this gap by presenting coherent, self-contained mathematical definitions for Scenario-Based vocabulary. Key terms such as scene, situation, concrete scenario, and logical scenario are defined within a mathematical framework proposed to model traffic constellations using the Behavioral modeling approach.

With the establishment of mathematical scenario definitions, the next step involves defining a generic framework for a dissimilarity metric for scenarios. This metric aids search and optimization algorithms in identifying a diverse range of driving scenarios, thereby addressing the challenge of avoiding redundant scenario selection for ADS testing purposes.

The structure of this report is as follows:

- Chapter 2: Provides an overview of existing scenario definitions in the literature followed by description of Behavioral Modeling concepts within the context of traffic systems using an illustrative example.
- Chapter 3: Covers dissimilarity concepts, modeling choices, assumptions, and their formulation.
- Chapter 4: Focuses on the implementation of the proposed dissimilarity metric.
- Chapter 5: Concludes with perspectives on future work.

2 Scenario definitions and behavioral modeling

This section delves into the importance of scenario definition within the context of assessing the safety of automated vehicles. It conducts a literature survey to examine the current methods employed for scenario definition and identifies the existing gap in establishing a mathematical definition for scenarios. Subsequently, it introduces the Behavioral Modeling (BM) approach for mathematically modeling dynamical systems and explores its application in traffic modeling. To illustrate this approach, we demonstrate its implementation by modeling a traffic system as a dynamical system. Through this example, we elucidate the rationale behind the BM approach, outline its key concepts, and demonstrate their practical application.

2.1 Background

There have been many recent advancements in autonomous driving, leading to the deployment of vehicles with high levels of autonomy in the market (SAE level 3 and above) [1]. These vehicles increasingly shift the responsibility of the dynamic driving tasks from the human driver to the Automated Vehicle (AV). Demonstrating that the safety of these systems is at par with human drivers remains an open challenge and makes the testing and verification of AVs vital from a safety standpoint. Many existing techniques are employed for the safety verification of AVs such as on road testing, Scenario-Based Testing (SBT), shadow testing, and formal verification [2]. Among these, one of the most prevalent methods in the literature is Scenario-Based Testing. Scenario-based testing involves the evaluation of a system's performance by exposure to a set of scenarios that is representative of the system's operating environment. The environment and conditions in which the AV is expected to operate without any safety violations are termed the Operational Design Domain (ODD). Scenario-based testing aims to ensure safe operation in the entire operational domain of the AV, while simultaneously limiting the number of relevant test scenarios.

The ISO standard for Safety Of The Intended Functionality (SOTIF), which is a key standard for AV safety verification and validation, establishes Scenario-Based Testing as the state-of-the-art [3]. Furthermore, there is significant development effort for standardized safety verification and validation of AVs revolving around scenario-based approaches, such as the European Union-funded HEADSTART [4] and SUNRISE projects [5], the PEGASUS family of projects in Germany [6], and the SAKURA project in Japan [7]. There is also concerted effort for the development and maintenance of standardized scenario databases such as the Safety Pool [8], StreetWise [9], and ADScene [10] databases.

2.2 Literature review on scenario definition

With this growing spotlight on Scenario-Based Testing, there is a need for standardized notions of scenarios and related terminology for repeatable, reproducible, and transparent testing. These notions need to be defined in a manner that allows for clear communication of ideas between various stakeholders without any room for ambiguity. In addition, SBT typically involves the performance evaluation of the AV through the use of certain performance indicators or metrics, which requires a mathematical formulation of the test scenarios. This further underscores the need for definitions of scenario-related vocabulary that are mathematically sound and

consistent. There have been numerous attempts in literature, to arrive at standardized definitions of vocabulary relating to Scenario-Based Testing.

The ISO Standard 34501 was recently published to standardize Scenario-Based Testing-related vocabulary [11]. In addition, there exist several works in literature that define scenarios and related terminology [12]–[16]. Among these, the definition of a scenario by Ulbrich et al. [12] is the most well-known and widely accepted, which states that ”A scenario describes the temporal development between several scenes in a sequence of scenes. Every scenario starts with an initial scene. Actions & events as well as goals & values may be specified to characterize this temporal development in a scenario. Other than a scene, a scenario spans a certain amount of time.” Gelder et al. [16] review the definitions of a ’scenario’ available in the literature, along with their associated shortcomings and inconsistencies.

In addition to these definitions of scenarios, numerous standardized description languages such as OpenSCENARIO by ASAM [17], Scenario Description Language by WMG University of Warwick [18], and SCENIC by University of California Berkeley [19] have been developed in recent years. These scenario description languages are focused on describing scenarios in machine-readable formats, which can be used to perform simulations and recreate scenarios. However, they lack a mathematically consistent, standardized foundation that aligns with definitions in the literature. A functional scenario [20] described in abstract, semantic terms requires parsing to generate corresponding logical scenarios [21].

This generation process differs between parties and is missing a standardized mathematical approach as shown by Costa et al. [22] [23], Zhang et al. [18] and Menzel et al. [21]. Even with these standardized description languages, the resulting concrete scenarios obtained during simulation from the same functional scenario may differ between simulation platforms, again leading to ambiguity. Alnaser et al [24] provide mathematical definitions of a scenario and scene, however, they define a scenario at an instant in time, when the ego vehicle ”makes a decision”. This is contradictory to the widely accepted notion of a scenario as being over a time interval, and is also unclear as to what ”making a decision” entails. This would be difficult with testing complex black-box systems that make several decisions at each instant, the knowledge of which might be unavailable or even irrelevant for testing. Andreotti et al. [25] propose mathematical definitions for scenarios and related terminology based on the definitions proposed by Ulbrich et al [12]. They define a scenario to be comprised of a set of objects, and related functions. A scenario is defined from the point of view of a specific object at discrete time instants.

The evolution from one time instant to the next is solely specified by functions describing the motion path of the objects. In this definition, no information about the temporal evolution between the time instants is included. Representing a spatially stationary object with non-spatial variables, such as traffic light for computing Key Performance Indicators (KPIs) requires an ad-hoc extension. Thus, this scenario description is less suitable for objectively evaluating scenarios and computing KPIs. Furthermore, the framework requires the definition of functions for describing a scenario, which may not be relevant to the end goal of Scenario-Based Testing, i.e., evaluating scenarios. Moreover, for defining a ’situation’, Andreotti et al. suggest a ’relevant function’ that filters out ’irrelevant objects and information’ without explicit specification of what it entails which could lead to ambiguous interpretations.

In all the different works mentioned so far, numerous discrepancies exist in how the term 'scenario' is used, providing room for ambiguous interpretations of scenarios. The existing mathematical definitions are inadequate for addressing this issue. As pointed out in interviews by experts in the AV industry and research, there is a need for a scenario definition that can be used as a common basis for research on Scenario-Based Testing [26]. In particular, there is a need for precise mathematical definitions of scenario-related terminology, which lead to unambiguous interpretations of scenarios. In this deliverable, we address the gap in literature and present a novel way of mathematically defining scenarios, by viewing the entire traffic constellation and the surrounding environment as a dynamical system. Using the Behavioral Modeling (BM) approach by Willems et al. [27], we present a mathematically rigorous, adaptable framework to define scenario related vocabulary and ground the definitions in the existing literature on Scenario-Based Testing vocabulary.

2.3 Behavior Modeling Approach

The primary source leading to the use of the Behavioral approach is the article by Jan C. Willems [28], published in 2007. The inspiration behind the behavioral approach stems from recognizing that classical system-theoretic thinking is inadequate when addressing the fundamental principles that system theory aims to tackle, specifically, open and interconnected systems. An open and interconnected system interacts with its environment and consists of interacting sub-systems.

Classical system theory models include a system with inputs, outputs, and signal-flow graphs. Inputs are utilized to represent how the environment impacts the system, whereas outputs are employed to represent how the system influences the environment. Output-to-input assignments, such as series and feedback connections, represent interconnections between different components. A system is thus perceived as a medium that transmits and alters signals, transferring them from the input to the output channel. Interconnections are regarded as pathways facilitating the transfer of outputs from one system to act as inputs for another system.

The laws governing physical phenomena primarily establish relationships among system variables, whereas interconnection implies sharing variables among sub-systems. In the behavioral approach, a system's behavior is viewed as a holistic model where all variables are treated equally. After detailed analysis, the system variables are divided into input and output sets depending on the model's intended use. This division aligns with classical input/output models, which can be integrated into the behavioral framework with additional structure. The main goal of behavior modeling is to establish a mathematical language for handling models of open systems and their interactions while maintaining fidelity to the underlying physics. A driving scenario can thus also be perceived as a behavioral model, represented by the number of actors, road layout, environmental conditions, and the path traversed by actors in the environment.

Since this deliverable concentrates on computing a dissimilarity between driving scenarios, some terms must be clarified and defined. First, the Operational Design Domain (ODD) of ADS is the specific operating domain in which it is designed to function [29]. Defining an ODD is crucial because it clarifies the terms required to describe a driving scenario.

Adopting the behavioral modeling approach [28], these terms are defined as follows:

1) Time T: The independent variable of time, which is a subset of R (the set of real numbers). The real world is modeled as a continuous time system.

2) Variables of Interest (VOI) ζ: These are manifest variables that are of interest from a modeling point of view. In this case, they consist of (but are not limited to) positions, velocities, accelerations of actors, and traffic light states.

3) Signal space W: The space of all values the VOI can take.

4) Universum U: All possible mappings from T to W, represented as W^T.

5) Behavior B: A subset of U where some conditions, as outlined by the modeler are met. Example conditions include:

- Physical laws like Newton's laws.
- Control laws for stability and safe distance keeping.
- Laws defining goal-oriented motions. Behavioral relations: The above-mentioned conditions form behavioral relations when expressed as mathematical equations. Therefore, behavioral relations are all the mathematical equations that define the behavior of the dynamical system of interest. They may or may not be associated with a specific actor. They include VOI (ζ), latent variables ($\bar{\zeta}$), and parameters (η). These equations may include:
- Ordinary Differential Equations (ODEs)
- Partial Differential Equations (PDEs)
- Algebraic equations and inequalities.

6) System of Interest (SOI) Σ: The real-world dynamical system under consideration for modeling. For this deliverable, the system consists of actors, environments, and their interaction with each other in road-based traffic systems. It is defined as the tuple:

$$\Sigma = \{T, W, B\}$$

7) Latent variables $\bar{\zeta}$: The auxiliary variables that appear in the behavioral relations of a system Σ, apart from the VOI.

8) System parameters η: All constant quantities appearing in behavioral relations.

2.4 Scenario Terminology Definition

Using the definitions listed above, the notions of scenario from the literature [12], [20] are also re-defined:

1) Concrete scenario: A specific trajectory of the VOI that lies within the behavior of the system. Note that this is a vector trajectory. This definition of a scenario is in agreement with a concrete scenario defined by Menzel [20].

2) Logical scenario: The logical scenario is obtained by specifying the behavior B of the complete dynamical system in the form of a state-space model. The state space model includes the defined ranges of values for the parameters. This definition is in agreement with Menzel's [20] definition. If all the parameters of the logical

scenario are specified, then it uniquely defines a concrete scenario. A logical scenario contains one or more concrete scenarios. If behavior B is specified to contain only one concrete scenario, the behavior is both a concrete and a logical scenario.

3) Functional scenario: A functional scenario is defined as a semantic description of a scenario. Specifying a functional scenario helps to derive the boundaries of the systems of interest and subsequently derive the signal space W, time T, and universum U.

4) Scenario category: A subset of the behavior B. This subset is specified by characterizing concrete scenarios based on some common properties.

5) Objects O_i: A physical entity of interest from a modeling point of view, for example, a tree, road, or car. Each object is identified by a unique index i.

6) Variables: Quantities that are allowed to vary with time. They are divided into VOI and latent variables.

7) Actors: Any object O_i with associated variables. Hence, actors are dynamic elements. However, the VOI need not be associated with a single actor. This definition deviates from [9] as dynamic elements include traffic lights and changing road signs.

8) Scenery: Objects which are stationary with respect to time.

9) Scene: A scene is the set of values of all VOI taken at a given time. Hence, evaluating a concrete scenario (single trajectory) at the given instant is a scene. A scene does not include latent variables. A scene may or may not contain the system's full memory sufficient to simulate the system or predict future scenes. It depends on the VOI chosen and knowledge of the behavior B. This deviates from the definition of Ulbrich [12] as it does not contain the values of the system parameters or latent variables. Defining a driving scenario using the behavioral approach implicitly adds a level of distinction between driving scenarios. If a driving scenario universum has different behavioral relations, different Variables of Interest (VOI), or different objects when compared with another, it directly follows that the two driving scenarios under consideration are different. Similarly, within a driving scenario universum, there can exist multiple concrete scenario trajectories which can be similar/different. To understand how the similarity of such concrete scenario trajectories can be quantified, a sample SOI is modeled and used in the remainder of this report.

2.5 Modeling a system of interest

To model a system of interest (SOI) for driving scenarios, fundamental parameters such as the road configuration, number of actors (vehicles, pedestrians, scenery elements), and a region boundary must be selected. For this deliverable, an intersection road layout has been modeled as an example. The terms defined are in previous section then listed below:

1) Scenery: The boundary box considered is a square with $50\ m$ side length as shown in Figure 3. The width of each lane is $3.75\ m$ based on standard road dimensions.
2) Actors: The vehicles and the pedestrian are modeled as ellipses and circles respectively, to have an approximate mathematical representation in 2D.

3) Variables of Interest (VOI): The Variables of Interest (VOI) include the positions and velocities of the actors in the boundary box. They are:
   a. Vehicle 1: $(x_{v1}, y_{v1}) \in [0, 50]$ $(v_{xv1}, v_{yv1}) \in [-v_{max}, v_{max}]$
   b. Vehicle 2: $(x_{v2}, y_{v2}) \in [0, 50]$ $(v_{xv2}, v_{yv2}) \in [-v_{max}, v_{max}]$
   c. Pedestrian: $(x_{p1}, y_{p1}) \in [0, 50]$ $(v_{xp1}, v_{yp1}) \in [-v_{pmax}, v_{pmax}]$
   The notations (x_{v1}, y_{v1}), (x_{v2}, y_{v2}), and(x_{p1}, y_{p1}) represent the x and y positions of vehicle 1, vehicle 2, and the pedestrian, respectively. The notations (v_{xv1}, v_{yv1}), (v_{xv2}, v_{yv2}), and (v_{xp1}, v_{yp1}) represent the velocity in x and y directions for vehicle 1, vehicle 2, and pedestrian respectively. The notation v_{max} and v_{pmax} represents the maximum allowed velocity for actors and pedestrians in the system.
4) Behavior (B): The behavioral constraints and assumptions are:
   a. The velocity of the actors is equal to the rate of change of the position:
      i. $\dot{x}_{v1} = v_{xv1}, \dot{y}_{v1} = v_{yv1}$
      ii. $\dot{x}_{v2} = v_{xv2}, \dot{y}_{v2} = v_{yv2}$
      iii. $\dot{x}_{p1} = v_{xp1}, \dot{y}_{p1} = v_{yp1}$
   b. The actors, modeled as ellipses and circles, can not overlap at any time in the scenario.
   c. The vehicles are modeled to move freely along the road and not anywhere else in the scenery. This imposes additional constraints on the (x, y) positions of the vehicles.
   d. The pedestrian is free to move anywhere within the boundary box. A key point to note at this stage is that this chosen SOI and behavior is just an example to understand what could be considered similar between two scenario trajectories within an SOI. This report aims to highlight the importance of such a modeling approach, thus giving a set of guidelines to the designer for formulating an SOI for testing.

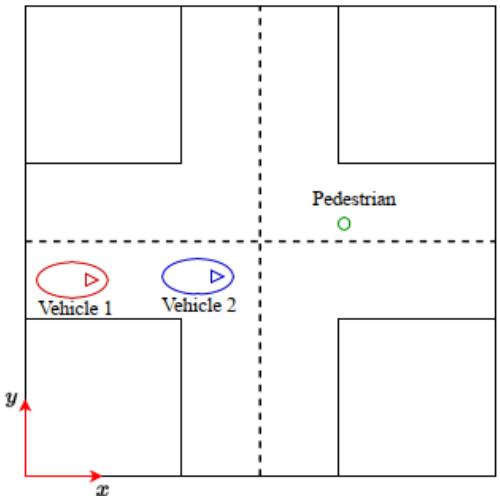


Figure 3 Modeled System of Interest (SOI) Σ

3 Dissimilarity metric

This chapter provides a quantitative measure that assesses the degree of resemblance, proximity, or correlation between different entities. It conducts a literature survey to examine the current methods employed for dissimilarity of scenarios. Subsequently, it provides an overview on metric foundation and finally addresses the formulation of the dissimilarity metric for computing the difference between two given concrete scenarios.

3.1 Literature review

To formulate a quantitative measure for assessing the similarity/dissimilarity between two driving scenarios, it is crucial to understand what is characterized as similar/dissimilar. As a human observer, one can consider multiple features to call a set of driving scenarios similar. These features could include (but are not limited to):

1) Position and orientation of vehicles

2) Speed and acceleration of vehicles

3) Time duration of scenarios

4) Obstacles in the environment

5) Driver intention and behavior

6) Road type and weather conditions

7) Vehicle interactions and spatial awareness

8) Collisions due to high traffic density

At this stage, given any two driving scenarios, the choice for characterizing them as similar or different becomes a subjective matter based on the features considered. Including as many features as possible increases the problem's complexity significantly.

To address this challenge, researchers delve into metrics that can differentiate between scenarios based on a simplified and finite set of features. Su et al. [30], Sousa et al. [31], and Tao et al. [32] present surveys on existing measures of similarity in literature. The most commonly used measures from the surveys are Euclidean distance, Lock-step Euclidean distance (LSED), Dynamic Time Warping, Longest Common Sub Sequence (LCSS), Edit distance (EDR), Edit Distance with Real Penalty (ERP), Normalized Euclidean distance (NED), Fr´echet distance (FD) and Hausdorff distance. Numerous researchers have employed these metrics across diverse applications and are subject to individual preferences.

Duchene et al. [33] in their work, compute the similarity measure for heterogeneous time-series data using the LCSS. Atev et al. [34] use the Hausdorff distance to cluster vehicle trajectories. Seyler et al. [35] use both Hausdorff and Frechet distances to compute path similarity for macro-molecular pathways. Weise et al. [36] also use the same measures as Seyler [35] in a multi-objective path-finding problem.

Apart from the above-mentioned measures, a few researchers propose new measures that aim at scenario dissimilarity. Kerber et al. [37] introduce an approach for scenario clustering through spatiotemporal (location and time-based) filtering of driving scenarios within a database. The database consists of driving data of a Vehicle Under Test (VUT) in the center lane of a three-lane highway. A total of eight regions around the VUT (hereby referred to as "slots") are considered at each scene (as defined in [12]) of a driving scenario. A custom distance measure is proposed, which first evaluates the distance between corresponding scenes of each scenario, yielding a sequence of distances. Finally, this sequence is mapped to a single value indicating scenario distance. The scene distance is computed by calculating the longitudinal distances between the traffic objects in all eight slots. The empty slots are ignored, and the slots with traffic objects present in one of the two scenes are labeled as 'vacant' and are penalized. The approach proposed by the authors is interesting for highway driving scenarios, but it is not generic and cannot be applied to other driving scenarios.

Bernard et al. [38] presented a trajectory clustering approach for scenario-based testing of ADS. In their definition, a trajectory is a sequence of vehicle states measured at discrete regions in the road network. Each trajectory is assigned an ID and converted into a histogram representation that captures the time distribution across different regions.

The dissimilarity between two normalized histograms is then calculated using the chi-squared distance, reflecting differences in the time spent by the vehicle at various regions. The chi-square distance is usually applied to applications that test how well a model compares to actual observed data. It requires that the data is random, mutually exclusive, and drawn from a large sample of independent variables. Although this measure helps analyze differences between categorical data (like histograms), it is sensitive to the number of samples. The approach's sensitivity to sample size can lead to distances seeming significant even when they are not, thus representing a limitation of the method. Nie et al. [39] propose a three-step trajectory similarity framework for Automatic Identification System (AIS) data. First, trajectory segments are extracted based on the driving data. The trajectory segment is then mapped to the corresponding grid cells according to spatial position. Second, directional relationships are assigned, the corresponding neighborhood is assessed, and a weighing factor is chosen. Last, trajectory similarity is determined by comparing the number and proportion of common cells in representative trajectories.

The proposed trajectory similarity analysis method solely considers spatial position input, neglecting the inclusion of temporal (time-based) input. This exclusion indicates a constraint inherent in the approach. The concept of scenario dissimilarity in the context of novelty assessment is introduced in Jonkman's master report [40]. The author first represents a scene by the position and orientation of each vehicle, their distance, and two angles to address relative positions and orientations. Using this scene representation, a dissimilarity measure for scenes is constructed based on the Euclidean distance between representing quantities. Finally, the scenario dissimilarity is obtained by computing the average distance between the corresponding scenes of each scenario. The author assumes equal time intervals between scenes for all compared scenarios and that they start at the same time stamp. This dissimilarity formulation additionally mandates that the two compared scenarios must possess an equal count of scenes.

Collectively, Jonkman's approach results in a measure of scenario dissimilarity and not a metric. The primary emphasis of the approaches by [37]–[40] is computing dissimilarity based on a scenario trajectory-level formulation. While these approaches serve as a source of inspiration for computing dissimilarity, their

utilization of scenario trajectories and a limited set of features restricts their generality. Considering the vital role of scenario criticality in testing Automated Driving Systems (ADS), it is possible to compute dissimilarity specifically at the most critical scene, rather than across the entire scenario. In conclusion, to the best of the authors' knowledge, there are currently no methods that specifically concentrate on formulating a generic similarity/dissimilarity metric based on the most critical scene. This particular aspect will be the subject of investigation in this report.

3.2 Metric Foundation

A metric for dissimilarity plays a pivotal role in comparing and evaluating various entities, systems, or phenomena. It is a quantitative measure that assesses the degree of resemblance, proximity, or correlation between different entities [41]. A standardized and objective approach to quantifying dissimilarity forms the basis for making informed decisions and drawing meaningful conclusions. This section discusses the concept of dissimilarity and the factors to consider for formulating a metric for differentiating between driving scenarios.

3.2.1 Concept of Dissimilarity

Using the definitions and SOI formulated in Section IV, a value of dissimilarity indicates the difference between any two concrete scenarios within each SOI. It is a mapping from the product of the behavior space $B \times B$ to a value in R_0^+ (the set of positive real numbers including zero), depending on the features considered. These features can include position and orientation, speed and acceleration of vehicles, time delays for the actors, and more, depending on the requirement of the test designer.

In the SOI considered, the features are limited to the positions and orientation, which are derived based on the velocity of the actors. To differentiate between two scenarios within the SOI, computing a difference between position and orientation should fully represent the dissimilarity. However, this need not always be the case. Consider an example of two scenarios where two vehicles are in a car-following trajectory in the straight section of the SOI shown in Figure 3. Assume that both these scenarios involve a collision between the two vehicles occurring at different locations on the straight road. Mathematically, both scenarios are different based on the length of the trajectories in each scenario and the point of collision with respect to global positioning. However, one would intuitively expect that these two scenarios considered here are similar. Evaluating the dissimilarity between these two scenarios, for instance, by calculating the distance between the points of collision will not be representative. To deal with such cases in the scenario space, a second level of abstraction is introduced in the SOI by dividing the road network into a mesh grid.

3.2.2 Abstraction with a Mesh grid

The scenario space is a search space with multiple dimensions based on the VOI. Calculating a dissimilarity at this level of complexity becomes computationally expensive. The complexity versus computation trade-off is fundamental in many research areas, especially computer science, mathematics, and algorithm design. In this specific application of formulating a dissimilarity metric, one benefits by reducing the system complexity by abstraction to differentiate between concrete scenarios in a scenario space. Thus, dividing a system of interest into different areas using a mesh grid can simplify the evaluation of dissimilarity while also addressing

the example mentioned in the previous subsection. Consequently, the assessment of dissimilarity can exclusively use the specific grid cells that impact the scenario. A sample method for meshing the modeled SOI is shown in Figure 4, thus dividing the road network into twelve grid cells. This meshing of the scenario space adds the first level of abstraction to the dissimilarity evaluation: Two scenarios where the actor's trajectories lay on different grid cells can already be considered distinct. At this stage, one can ask how to choose appropriate meshes. There can be multiple ways in which the road network in the modeled SOI can be divided into meshes. The choice of selecting a particular mesh grid is subjective, like the choice of features for evaluating dissimilarity. The mesh grid selected will therefore be unique to the designer. This report provides one possible set of guidelines to the designer to choose a mesh grid for any road network in an SOI. The guidelines are:

1) The width of each grid cell is equal to the lane width under consideration.

2) Each lane is one grid cell until:

a) The end of the lane, as per lane markings.

b) The lane splits into two to more lanes, where each split lane is a new grid cell.

c) The lane joins a second lane or a junction. For example, lanes merging into a highway or intersection.

d) A pedestrian/bicycle crossing splits a lane. In this case, it divides the lane into three grid cells, including the crossing.

3) The presence of objects in scenery (such as traffic lights and signboards) splits a lane into two grid cells.

4) Change in road curvature over a threshold results in splitting the lane into two or more grid cells based on the designer's requirement. In formulating these guidelines, a mindful approach is taken to accommodate the potential inclusion of real-world road layouts, lane markings, and scenery objects.

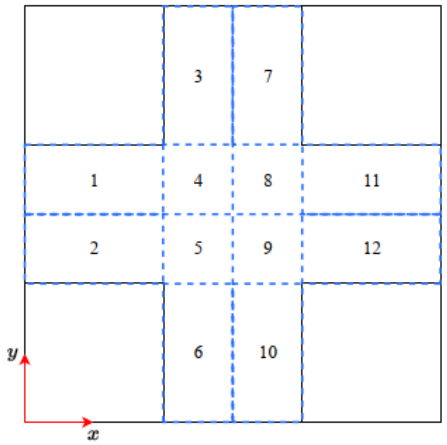


Figure 4 A sample representation of meshing

3.2.3 Dissimilarity at Critical Scene

The concept of dissimilarity and criticality in driving scenarios are crucial in shaping the decision-making process and ensuring road safety. As explained in Section V-A, dissimilarity refers to the perceived resemblance between any two scenarios and depends on the viewer's discretion. On the other hand, criticality pertains to the level of urgency and importance assigned to a specific driving scenario. Certain situations demand immediate attention and decisive actions to prevent accidents or fatal outcomes. Criticality assessment involves evaluating factors such as the speed of oncoming vehicles, the proximity of pedestrians, road conditions, and weather, among others. Mathematically, the criticality function defined in this study maps from the behavior space B to a value in R, quantifying how unsafe a given concrete scenario is. Understanding that dissimilarity and criticality are distinct but relevant, makes them pivotal for comparing and analyzing concrete scenarios.

While selecting scenarios for testing ADS, choosing the most critical scenarios can confirm that the ADS operates safely. From a broad perspective, the utmost critical scenario arises from the combination of collision proximity and severity. This combination of collision proximity and severity can be reduced to a scene in the scenario known as the critical scene. Thus, the dissimilarity between any two scenarios can be evaluated based on differences at the most critical scenes of each scenario. This simplification for the dissimilarity assessment reduces the problem's domain from trajectory to point-level analysis. Using a single scene for evaluating dissimilarity simplifies the complexity of the analysis.

3.2.4 Assumptions for the dissimilarity metric

Based on the deliverable consolidated so far, considering the behavioral modeling approach and the definition of mesh grid, the key hypotheses and pre-conditions on which the formulation of a dissimilarity metric lies is outlined below:

1) Two scenarios can only be compared if they lie within the same operational design domain (behavior space). Scenarios in a different behavior space are already considered 100% dissimilar.

2) Two scenarios can only be compared if all the actors' trajectories lay on the same sequence of grid cells. If the sequence of grid cells differs between two scenarios, they are considered 100% dissimilar.

3) The difference between the two scenarios will be evaluated by comparing features at the most critical scene in each scenario and not the entire scenario.

4) The vehicle under test cannot collide with more than one actor at any instant. This situation is assumed to be rare and complicates the computations for the dissimilarity; hence it is omitted. These assumptions define the boundaries and constraints for the dissimilarity metric definition and also acknowledge potential dependencies in the modeling approach followed.

3.2.5 Properties of the dissimilarity metric

As stated in previous section, the dissimilarity is a representation of how different two concrete scenarios are. The resulting dissimilarity value will be positive after comparing two concrete scenarios based on a feature set. Since dissimilarity is a metric for comparing two scenarios, the result can either be binary (0% dissimilar

or 100% dissimilar) or continuous and lie within the 0% and 100% bounds. In this report, 0% and 100% dissimilarity between two concrete scenarios are equivalently represented by 0 and 1. This $0-1$ normalization ensures simplicity and gives an intuitive understanding of dissimilarity [37]. Another advantage of following this scaling is that it is helpful for future researchers to use the method proposed for dissimilarity calculation in machine-learning algorithms and search algorithms like gradient descent-based optimization for finding diverse critical driving scenarios. Normalizing the output from the dissimilarity calculations can lead to faster convergence in search algorithms [42]. Let d denote the value for dissimilarity. The dissimilarity calculation should satisfy the fundamental mathematical properties of comparison metrics [41]. The general mathematical properties that any metric should satisfy are:

1) *Non-negativity*: The distance d between any two points in the metric space should always be non-negative. In other words, the distance is a non-negative real number:

$$d(x,y) \geq 0$$

2) Identity of indiscernible: The distance between two points is zero if and only if the points are the same:

$$d(x,y) = 0 \iff x = y$$

3) Symmetry: The distance between two points is the same regardless of the order in which they are considered:

$$d(x,y) = d(y,x)$$

4) Triangle Property: The triangle inequality property states that for any three points x, y, and z in the metric space, the sum of the distances between these points is always greater than or equal to the distance between any two of those points. Mathematically, it can be expressed as:

$$d(x,y) + d(y,z) \geq d(x,z)$$

$$\forall (x,y,z) \; in\ the\ metric\ space$$

The validation of whether the dissimilarity d will fulfill all the properties depends on the approach by which d is calculated. Based on the fulfillment of these properties, the dissimilarity can either be a true metric or categorize into one of the following types [41]:

1) Pseudometric: The identity of indiscernible has an exception, where:

$$d(x,y) = 0 \; for\ some\ x \neq y$$

2) Quasimetric: All axioms satisfied except the symmetry property.

3) Semimetric: All axioms satisfied except the triangle property.

4) Premetric: Relaxing the last three axioms leads to the notion of premetric. This is not a standard type.

5) Pseudoquasimetric/Hemimetric: Relaxes both the indiscernibles axiom and the symmetry axiom, and is simply a premetric satisfying the triangle property. Building upon these fundamental concepts,

assumptions, and properties, the subsequent section explains how the dissimilarity metric calculation is performed.

3.3 Dissimilarity metric formulation

This section addresses the formulation of the dissimilarity metric for computing the difference between two given concrete scenarios. It begins by explaining the process of calculating the critical scene from a provided scenario. Subsequently, it introduces the notion of discrete and continuous variables, highlighting their significance in this context. Last, the step-by-step process of calculating the dissimilarity using the discrete and continuous variables is explained.

3.3.1 Critical Scene Calculation

To evaluate the dissimilarity d between two concrete scenarios in the modeled SOI, the difference in the variables of interest is to be evaluated at the scene of maximum criticality. In the 2D modeling approach for the scenario trajectories, the variables of interest for dissimilarity, which can fully represent the actors involved in a maximum criticality condition, are the position (x, y) and orientation (θ) in the global frame of reference. To compare the VOI at the scene of maximum criticality for dissimilarity computation, it is first essential to understand what is critical and how to calculate it mathematically.

One of the most basic and computationally inexpensive methods to evaluate criticality is proximity to the objects in the SOI. Mathematically, this corresponds to the minimum distance between all objects. Using the minimum distance as a criticality function, each scenario can be reduced to one scene where the system under test (an actor) will either have a collision or a near-miss condition. The pseudo-code for this formulation of the criticality function is shown in Algorithm 1.

Algorithm 1: Pseudo-code of the Criticality function

```
Data: Initial conditions for all actors in a scenario
Result: VOI corresponding to the minimum distance
        scene
ContinuationFlag ← True;
MinimumDistance ← Inf ;   /* Initially set
 to an arbitrary maximum */
SceneCounter ← 1 ;    /* A counter to run
 through the loop */
NumberOfScenes ; /* Initialized to total
 number of scenes */
while ContinuationFlag do
    if SceneCounter ≥ NumberOfScenes then
        Store and Return VOI at MinimumDistance;
        ContinuationFlag ← False;
    else
        Compute VOI based on 𝔅, SceneCounter;
        Distances ← Computed distance between all
          actors;
        MinimumDistance ←
          min(MinimumDistance, Distances);
    end
    SceneCounter = SceneCounter + 1;
end
```

3.3.2 Discrete and Continuous Variables

Discrete and continuous variables are fundamental concepts in statistics and mathematics that categorize data types based on their nature. Discrete variables assume distinct, separate values and are often counted or enumerated, while continuous variables can take on any value within a certain range and are typically measured [43].

In the context of evaluating dissimilarity between the most critical scenes of two scenarios, it is important to record the variables that represent the critical scenes. Based on the critical scene calculation as the scene of minimum distance, the (x, y, θ) of the actors are returned as an output from the criticality function. For the modeled SOI, the outputs from the criticality function will be the VOI $(x_{v1}, y_{v1}, \theta_{v1}), (x_{v2}, y_{v2}, \theta_{v2})$, and $(x_{p1}, y_{p1}, \theta_{p1})$. Consider two scenarios, represented by these outputs $(x, y, \theta)_1$ and $(x, y, \theta)_2$, as inputs to compute the dissimilarity. Assuming vehicle 1 as the test vehicle, the post- processing steps of the inputs, thus computing a value for dissimilarity are explained below. Based on the positions returned at the event of maximum criticality, the minimum distance can occur between only two actors in the scenario. In this case, it can be with the second vehicle or the pedestrian. This understanding introduces the first discrete variable: $Actor\ type$ at the minimum distance $ActorType_c$. This variable belongs to a finite set of all combinations of actor interactions concerning the test vehicle. This finite set for the SOI considered is then: {'$Vehicle - Vehicle$','$Vehicle - Pedestrian$'}. The difference between $ActorType_c$ of two scenarios is then calculated as:

$$\Delta ActorType_c = \begin{cases} 0 & if\ (ActorType_c)_1 = (ActorType_c)_2 \\ 1 & otherwise \end{cases}$$

The goal of introducing the discrete variables helps in adding the third level of abstraction. Logically, two scenarios that result in $\Delta ActorType_c$ returning 1, indicate dissimilarity $d = 1$. Both of those scenarios are hence important with respect to testing and need to be selected. If $\Delta ActorType_c$ returns 0, further analysis is needed for dissimilarity.

Another discrete variable, based on the location of the test vehicle in the mesh grid, is also introduced: $GridCell_c$. The difference in the grid cell number between the two scenarios, at the point of minimum distance, is calculated as:

$$\Delta GridCell_c = \begin{cases} 0 & if\ (GridCell_c)_1 = (GridCell_c)_2 \\ 1 & otherwise \end{cases}$$

This discrete variable implicitly considers the differences in the positions of the actors: (x_{v1}, y_{v1}) with respect to $(x_{v2}, y_{v2})\ or\ (x_{p1}, y_{p1})$, based on the grid cell where they lie. Similar to the actor type at minimum distance, returns 1 if the grid cell of the minimum distance point with respect to the vehicle-under-test between the two scenarios is different, thus reflecting dissimilarity $d = 1$. If $\Delta GridCell_c$ returns 0, further analysis is needed for dissimilarity. The highlight at this stage is the option for the designer to add more discrete variables based on their intent for dissimilarity. Next, the notion of continuous variables is introduced for the angle outputs obtained from the criticality function at the point of minimum distance. Two angles are derived at the point of minimum distance: the relative heading angle θ_{rel} and the potential collision angle ϕ_c in the test vehicle frame.

The physical representation of the two angles in a critical scene is shown in Figure 5. These angles are pivotal in differentiating between critical scenarios due to their influence on the outcome of criticality. The θ_{rel} defines the orientation difference between two actors, providing a glimpse into how they are approaching each other. The ϕ_c indicates the point of minimum distance between actors and the dynamics of a possible collision. For each scenario, θ_{rel} is calculated as the angle between the heading vectors of the two actors at the minimum distance. The angle ϕ_c is calculated as the angle between the heading vector of the test vehicle and the vector connecting the geometric center of the test vehicle to the point on the test vehicle nearest to the second actor.

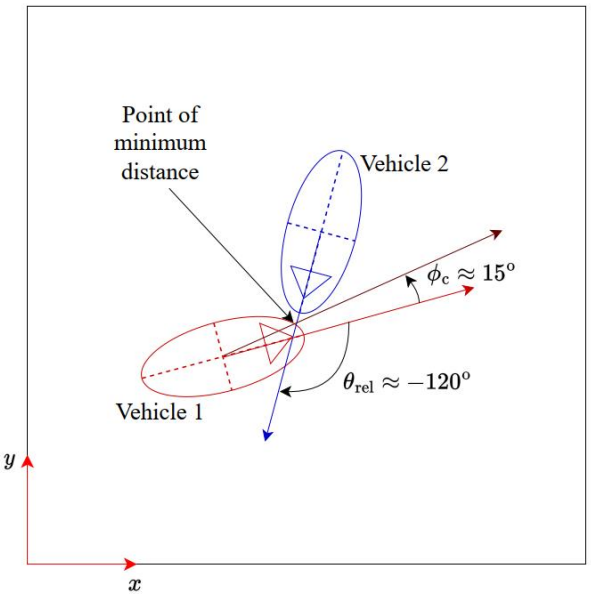


Figure 5 Representation of Relative heading θ_{rel} and angle of minimum distance in the local frame of reference ϕ_c. Counter-clockwise measurement is considered positive, and clockwise measurement is considered negative.

There exist multiple methods to compute the difference between angles. Based on the described properties, the expectation from the computed difference, based on all features, should lie within the $0-1$ normalization range. One possible way to achieve this is using the cosine similarity index [44]. The cosine similarity returns values in the range [0, 2], which can be reduced to $[0, 1]$ by multiplying the result by half. The cosine difference of the angles θ_{rel} and ϕ_c between two scenarios, denoted by ΔH and ΔC respectively, are then calculated as:

$$\Delta H = \frac{[1 - cos(\theta_{rel1} - \theta_{rel2})]}{2}$$

$$\Delta C = \frac{[1 - cos(\phi_{c1} - \phi_{c2})]}{2}$$

It is important to note that using cosine similarity is only one of the possible choices, that defines the value of dissimilarity. Using a different comparison metric will result in different values.

3.3.3 Dissimilarity Calculation

Having introduced two discrete and two continuous variables, the next step involves mapping them into a single value representing the dissimilarity. The mapping has to happen at two levels because the discrete variables can be $0\ or\ 1$, and the continuous variables have values between $0\ and\ 1$. The continuous variables are only considered for dissimilarity if the discrete variables have 0 as an output. The dissimilarity between two scenarios can be grouped into a single algorithm and are detailed in the flowchart, as shown in Figure 6. Implicitly considered is the highest level of dissimilarity, where the same sequence of mesh grid cells is essential for comparison; otherwise, it results in $d = 1$. To implement the formulated dissimilarity metric and evaluate its effectiveness, manual formulation of sample scenario trajectories is done within the modeled SOI and discussed in the next section.

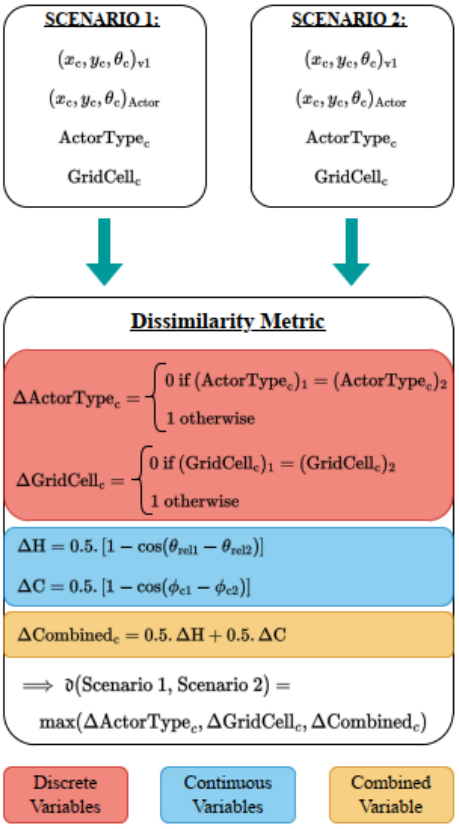


Figure 6 Flowchart for the dissimilarity metric proposed, with averaging by addition for the continuous variables

4 Dissimilarity metric implementation

This section presents the practical application of the dissimilarity metric on sample scenarios within the modeled SOI. Additionally, the utilization of this metric for forming scenario clusters using a clustering algorithm is discussed.

4.1 Dissimilarity for the modeled SOI

In the modeled SOI, with the intersection type of road junction, one possible mapping of scenario trajectories, with the paths for the three actors is depicted in Figure 7. Vehicle 1 (vehicle under test) is modeled to start its trajectory from the left side of the intersection, followed by executing a left turn. The sequence of grid cells that form the path for vehicle 1 is $(2, 4, 9, 8, 7)$. In contrast, the second vehicle (vehicle 2) also makes a left turn, but it initiates the maneuver from the opposite lane and the opposing side of vehicle 1. For vehicle 2, the grid cell sequence of the path is $(11, 8, 4, 5, 6)$. Additionally, the pedestrian's simulated movement involves crossing the road and passing through grid cells 7 and 3. Within one set of grid cell sequences of the actors, there can exist potentially different driving scenarios, based on varying initial velocities and time delays of the actors. Here, time delay refers to the difference in instances at which the actors start moving in the planned path. It is vital to note here that using a grid cell sequence also allows for lateral path deviations within each grid cell, thus still entertaining new possibilities to arise. It is also understood at this stage that a different set of paths for the actors will result in a different set of mesh grid sequences, thus resulting in an entirely different scenario. Different initial velocities and time delays for the actors within the considered grid cell sequence can result in different collision modes. Based on the three path intersections between the actors, each uniquely identified by grid cell numbers, a set of possible collision modes are shown in Figure 8.

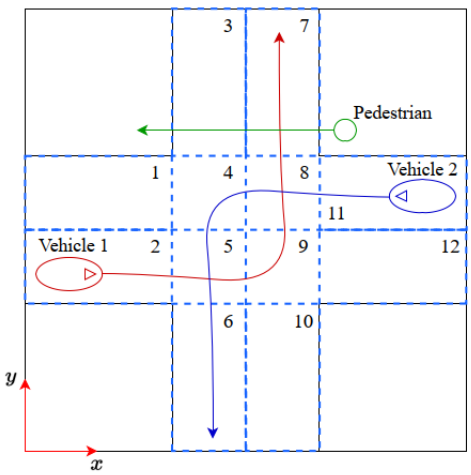


Figure 7 Base scenario trajectories for all actors in the modeled SOI

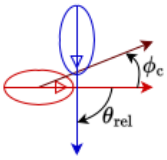
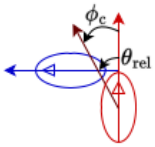
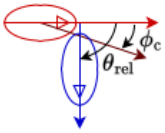
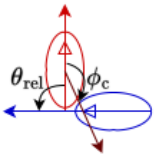
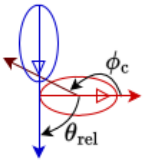
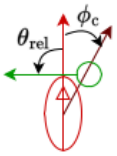
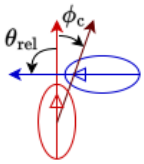
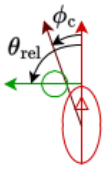
No.	Collision Modes	Values	No.	Collision Modes	Values
1		$\theta_{rel} = -90°$ $\phi_c \approx 40°$ ActorType_c = 'Vehicle-Vehicle' $\text{GridCell}_c = 5$	5		$\theta_{rel} = 90°$ $\phi_c \approx 40°$ ActorType_c = 'Vehicle-Vehicle' $\text{GridCell}_c = 8$
2		$\theta_{rel} = -90°$ $\phi_c \approx -30°$ ActorType_c = 'Vehicle-Vehicle' $\text{GridCell}_c = 5$	6		$\theta_{rel} = 90°$ $\phi_c \approx -160°$ ActorType_c = 'Vehicle-Vehicle' $\text{GridCell}_c = 8$
3		$\theta_{rel} = -90°$ $\phi_c \approx 160°$ ActorType_c = 'Vehicle-Vehicle' $\text{GridCell}_c = 5$	7		$\theta_{rel} = 90°$ $\phi_c \approx -40°$ ActorType_c = 'Vehicle-Pedestrian' $\text{GridCell}_c = 7$
4		$\theta_{rel} = 90°$ $\phi_c \approx -30°$ ActorType_c = 'Vehicle-Vehicle' $\text{GridCell}_c = 8$	8		$\theta_{rel} = 90°$ $\phi_c \approx 30°$ ActorType_c = 'Vehicle-Pedestrian' $\text{GridCell}_c = 7$

Figure 8 Collision modes for scenario path in Figure 7

4.2 Clustering scenarios using dissimilarity

The dissimilarity metric implementation in evaluating the difference between two given scenarios is discussed in the previous subsection. As the purpose of the dissimilarity metric is to ensure diversity by preventing redundant scenarios, an example of clustering a set of scenarios using a clustering algorithm (with the proposed dissimilarity metric as the distance function) is additionally explored in this section. The pedestrian is removed from the scenario space to simplify the modeled SOI, making it easier to implement and visualize the clusters. Using the same base path for the two vehicles from Fig. 6, the minimum distance and the angles (relative heading θ_{rel}, angle ϕ_c at the point of minimum distance corresponding to vehicle 1 frame) as a function of the velocity ratio of the actors is shown in Figure 9. The velocity ratio is sampled with 200 points, forming 200 scenarios, with each set of the two angles representing a scenario. The actors are modeled to start moving from their initial position at the start of the simulation time without any delays. It can be seen from the figure that there are two velocity ratio zones where the minimum distance is zero, which indicates a collision. The first zone occurs at $v1/v2 < 1$, which indicates that Vehicle 1 is slower than Vehicle 2. This can occur at the grid cell 5 in the road layout. The second zone at $v1/v2 > 1$ indicates the vice versa and

can occur at grid cell 8 in the road layout. When the velocity ratio is 1, the two vehicles stay clear of each other in their respective paths as a near-miss condition. The relative heading angle θ_{rel}, as expected, lies between $-90°$ and $+90°$. The angle at the minimum distance in the test vehicle frame ϕ_c is initially positive, indicating the minimum distance is at the left side of the test vehicle. The ϕ_c is then negative, indicating the minimum distance is at the right side of the test vehicle.

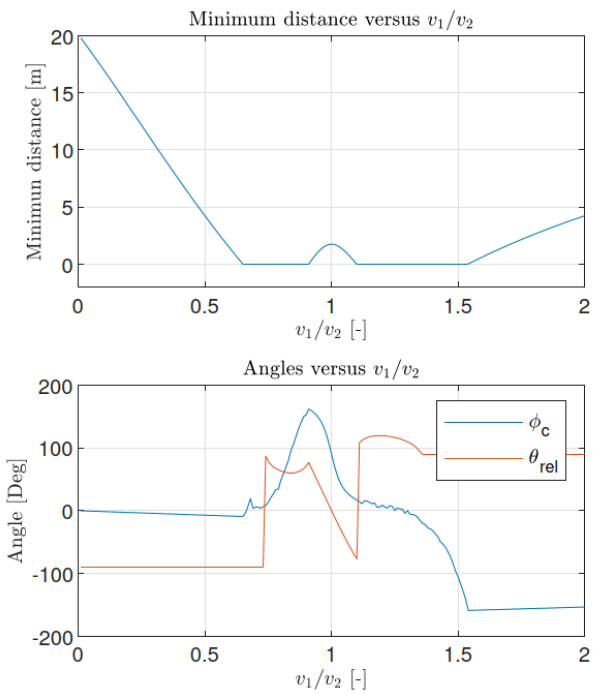


Figure 9 Minimum distance and continuous variables (θ_{rel}, ϕ_c) as a function of velocity ratio of actors

According to the path defined for the test vehicle, the minimum distance scene can occur at the grid cell numbers $\{2, 5, 7, 8, 9\}$. As per the definition of the dissimilarity metric, the occurrence of the minimum distance at a different grid cell between two scenarios results in 100% dissimilarity ($d = 1$). Hence there will be a minimum of five scenarios with a dissimilarity of 1, resulting in at least five distinct the five grid cell numbers of the test vehicle. It can also be seen from this figure that there can be more than five clusters in reality. For example, the points at grid cell 5 can be divided into at least two clusters: one at positive angles of θ_{rel} ($\approx 40°$ to $75°$) and one at negative angles ($\theta_{rel} \approx -90°$). To cluster the scenarios with the formulated dissimilarity metric, k-medoids clustering is applied. This is one of the possible methods that allow a custom distance while clustering. This type of clustering requires a priori choosing the number of clusters k. The process of clustering is conducted iteratively for various values of k, and the suitability of the selected number of clusters is assessed by utilizing the silhouette metric [45]. The silhouette metric is defined as a measure of how similar an object is to its cluster (cohesion) compared to other clusters (separation). Mathematically, it is expressed as:

$$s(i) = \frac{b(i) - a(i)}{max\{a(i), b(i)\}}$$

Here, $s(i)$ is the silhouette score for data point $i, a(i)$ is the average distance between data point i and other data points within the same cluster, and $b(i)$ is the average distance between data point i and all data points in other clusters. The silhouette score ranges from $-1\ to\ +1$, where a high value indicates that the object is well-matched to its cluster and poorly matched to neighboring clusters. The clustering configuration is appropriate if most objects have a high value. If many points have a low or negative value, the clustering configuration may have too many or too few clusters. An advantage of the silhouette metric is that it can be applied to any distance metric, making it easy to implement with the dissimilarity metric.

The k value of 8 is then selected by evaluating the highest average silhouette score for clustering values of k ranging between 5 and 15. The result of clustering the scenarios with $k = 8$ is shown in Figure 10. The clustering of scenarios with the dissimilarity metric indicates the possibility of using the metric in selecting diverse scenarios, thus considerably reducing the number of scenarios from $200\ to\ 8$, in the example given.

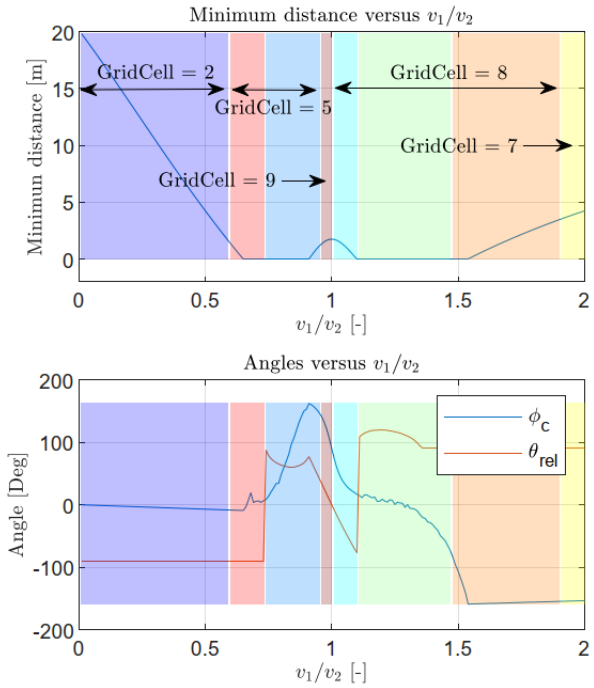


Figure 10 Minimum distance and continuous variables (θrel, ϕc) as a function of velocity ratio of actors, with clustering

5 Conclusions and follow-up

The objective of SymAware project is to develop a framework for awareness in multi-agent systems that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and humans.

The main goal of this deliverable is to ensure diversity within a scenario generation framework for Automated Driving Systems (ADS) testing and verification by preventing the creation of redundant scenarios and therefore targets to find unknown scenarios. To address this challenge, a generic framework to formulate a dissimilarity metric is presented to differentiate between driving scenarios accurately, thus preventing redundancies.

The framework first includes a behavioral modeling approach to define concrete driving scenarios, formally defining all the objects and attributes within the scenario. Next, the scenario space is split into a mesh grid for simplification, thus introducing a first level of abstraction. A second level of abstraction is then formulated by introducing the notion of discrete and continuous variables, derived based on a criticality function. The criticality function considered in this study is the minimum distance between all the actors in the scenario.

The dissimilarity metric between any two scenarios within the modeled space is then formulated by mapping the extracted discrete and continuous variables into a single value. The mapping is executed at two levels: mapping all continuous variables into a single value and then mapping this output with the discrete variables. The value from both mapping sequences finally results in the dissimilarity metric, signifying the difference between the two concrete scenarios.

In conclusion, this deliverable presents a generic framework for formulating a dissimilarity metric to differentiate between driving scenarios. Through meticulous research and analysis, it uncovers key insights on scenario modeling and dissimilarity. The primary findings include:

1) The behavioral modeling approach proves to be an effective method to formalize a generic framework for defining driving scenarios.

2) Abstraction of the road network in an SOI using mesh grids offers a clear visualization of the scenario, making it easier to perform spatial computations. Mesh grids can also be tailored to accommodate various resolutions, making them adaptable to different requirements.

3) The dissimilarity is determined based on the scene with maximum criticality between scenarios. This simplifies the dissimilarity analysis from trajectory-level to point level analysis.

4) Clustering of scenarios using the dissimilarity metric indicates the possibility of selecting non-redundant scenarios for testing Automated Driving Systems (ADS).

The output of this deliverable on the mathematical scenario modeling, and the proposed dissimilarity metric later will be used in the optimization algorithm in WP5 Task 5.2.2 to extract unknown unsafe scenarios of

SOTIF standard in the automotive use case. The generated scenarios will be used in WP5 Task 5.2.3 to assesses the effectiveness of increased awareness in automated driving functionalities.

Bibliography

[1] "Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles," SAE International, Standard, Apr. 2021.

[2] S. Riedmaier, T. Ponn, D. Ludwig, B. Schick, and F. Diermeyer, "Survey on scenario-based safety assessment of automated vehicles," IEEE Access, vol. 8, pp. 87 456–87 477, 2020.

[3] "Road vehicles - Safety of the intended functionality," International Organization for Standardization, Geneva, CH, Standard, Jun. 2022.

[4] "HEADSTART - Harmonised European Solutions for Testing Automated Road Transport ," https://www.headstart-project.eu/, [Accessed 23-11-2023].

[5] "Sunrise Project — Developing and providing a harmonized and scalable CCAM Safety Assurance Framework - ccam-sunrise-project.eu," https://ccam-sunrise-project.eu/, [Accessed 23-11-2023].

[6] "Pegasus Research Project - Securing Automated Driving Effectively," https://www.pegasusprojekt.de/en, [Accessed 23-11-2023].

[7] "SAKURA Research Project," https://www.sakura-prj.go.jp/, [Accessed 23-11-2023].

[8] "Safety Pool - Powered by Deepen AI and WMG University of Warwick," https://www.safetypool.ai/, [Accessed 24-11-2023].

[9] H. Elrofai, J.-P. Paardekooper, E. de Gelder, S. Kalisvaart, and O. Op den Camp, "StreetWise: scenario-based safety validation of connected and automated driving," TNO, Helmond, Tech. Rep., Jul. 2018.

[10] S. Geronimi and E. Arnoux, "ADScene - A path to a European scenarios database for ADS and ADAS specification, validation and homologation," https://www.vvm-projekt.de/fileadmin/user upload/Mid-Term/Presentations/VVM HZE EmmanuelArnoux.pdf , Mar. 2022, [Accessed 24-11-2023].

[11] "Road vehicles - Test scenarios for automated driving systems - Vocabulary," International Organization for Standardization, Geneva, CH, Standard, Oct. 2022.

[12] S. Ulbrich, T. Menzel, A. Reschka, F. Schuldt, and M. Maurer, "Defining and substantiating the terms scene, situation, and scenario for automated driving," in 2015 IEEE 18th International Conference on Intelligent Transportation Systems, 2015, pp. 982–988.

[13] J. Bach, S. Otten, and E. Sax, "Model based scenario specification for development and test of automated driving functions," in 2016 IEEE Intelligent Vehicles Symposium (IV), 2016, pp. 1149–1155.

[14] S. Geyer, M. Baltzer, B. Franz, S. Hakuli, M. Kauer, M. Kienle, S. Meier, T. Weißgerber, K. Bengler, R. Bruder, F. Flemisch, and H. Winner, "Concept and development of a unified ontology for generating test and use-case catalogues for assisted and automated vehicle guidance," IET Intelligent Transport Systems,

vol. 8, no. 3, pp. 183–189, 2014. [Online]. Available: https: //ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/iet-its.2012.0188

[15] E. Rocklage, H. Kraft, A. Karatas, and J. Seewig, "Automated scenario generation for regression testing of autonomous vehicles," in 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC), 2017, pp. 476–483.

[16] E. de Gelder, J.-P. Paardekooper, A. K. Saberi, H. Elrofai, O. Op den Camp, S. Kraines, J. Ploeg, and B. De Schutter, "Towards an ontology for scenario definition for the assessment of automated vehicles: An object-oriented framework," IEEE Transactions on Intelligent Vehicles, vol. 7, no. 2, pp. 300–314, 2022.

[17] "OpenSCENARIO," https://www.asam.net/standards/detail/openscenario/, 2022, [Accessed 24-11-2023].

[18] X. Zhang, S. Khastgir, and P. Jennings, "Scenario description language for automated driving systems: A two level abstraction approach," in 2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2020, pp. 973–980.

[19] D. Fremont, E. Kim, T. Dreossi, S. Ghosh, X. Yue, A. Vincentelli, and S. Seshia, "Scenic: a language for scenario specification and data generation," Machine Learning, vol. 112, pp. 1–45, 02 2022.

[20] T. Menzel, G. Bagschik, and M. Maurer, "Scenarios for development, test and validation of automated vehicles," in 2018 IEEE Intelligent Vehicles Symposium (IV). IEEE Press, 2018, p. 1821–1827. [Online].

Available: https://doi.org/10.1109/IVS.2018.8500406

[21] T. Menzel, G. Bagschik, L. Ise nsee, A. Schomburg, and M. Maurer, "From Functional to Logical Scenarios: Detailing a Keyword-Based Scenario Description for Execution in a Simulation Environment," 2019 IEEE Intelligent Vehicles Symposium (IV), pp. 2383–2390, Jun. 2019, conference Name: 2019 IEEE Intelligent Vehicles Symposium (IV) ISBN: 9781728105604 Place: Paris, France Publisher: IEEE. [Online]. Available: https://ieeexplore.ieee.org/document/8814099/

[22] A. A. Bruto da Costa, P. Irvine, X. Zhang, S. Khastgir, and P. Jennings, "Writing Accessible and Correct Test Scenarios for Automated Driving Systems," 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 1356–1363, Oct. 2022, conference Name: 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC) ISBN: 9781665452588 Place: Prague, Czech Republic Publisher: IEEE. [Online]. Available: https://ieeexplore.ieee.org/document/9945564/

[23] ——, "Translating Automated Vehicle Test Scenario Specifications Between Scenario Languages: Learnings and Challenges," 2022. [Online]. Available: https://www.semanticscholar.org/paper/ Translating-Automated-Vehicle-Test-Scenario-Between-Costa-Irvine/ 3cb72f80da9fb9dd03d6abbe83db28e27cf9b273

[24] A. J. Alnaser, M. I. Akbas, A. Sargolzaei, and R. Razdan, "Autonomous vehicles scenario testing framework and model of computation," SAE International Journal of Connected and Automated Vehicles, vol. 2, no. 4, pp. 205–218, dec 2019. [Online]. Available: https://doi.org/10.4271/12-02-04-0015

[25] E. Andreotti, P. Boyraz, and S. Selpi, "Mathematical definitions of scene and scenario for analysis of automated driving systems in mixed traffic simulations," IEEE Transactions on Intelligent Vehicles, vol. 6, no. 2, pp. 366–375, 2021.

[26] F. Beringhoff, J. Greenyer, C. Roesener, and M. Tichy, "Thirty-one challenges in testing automated vehicles: Interviews with experts from industry and research," in 2022 IEEE Intelligent Vehicles Symposium (IV), 2022, pp. 360–366.

[27] J. Polderman and J. Willems, Introduction to Mathematical Systems Theory: A Behavioral Approach, ser. Texts in Applied Mathematics. Springer, 1998.

[28] J. C. Willems, “The behavioral approach to open and interconnected systems,” IEEE Control Systems Magazine, vol. 27, no. 6, pp. 46–99, 2007.

[29] E. Thorn, S. C. Kimmel, and M. Chaka, “A Framework for Automated Driving System Testable Cases and Scenarios,” Department of Transportation, NHTSA, United States, Technical Report, Sep. 2018.

[30] H. Su, S. Liu, B. Zheng, X. Zhou, and K. Zheng, “A survey of trajectory distance measures and performance evaluation,” The VLDB Journal, vol. 29, no. 1, pp. 3–32, Jan. 2020. [Online]. Available:

http://link.springer.com/10.1007/s00778-019-00574-9

[31] R. S. D. Sousa, A. Boukerche, and A. A. F. Loureiro, “Vehicle Trajectory Similarity: Models, Methods, and Applications,” ACM Computing Surveys, vol. 53, no. 5, pp. 94:1–94:32, Sep. 2020. [Online]. Available: https://dl.acm.org/doi/10.1145/3406096

[32] Y. Tao, A. Both, R. I. Silveira, K. Buchin, S. Sijben, R. S. Purves, P. Laube, D. Peng, K. Toohey, and M. Duckham, “A comparative analysis of trajectory similarity measures,” GIScience & Remote Sensing, vol. 58, no. 5, pp. 643–669, Jul. 2021. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/

15481603.2021.1908927

[33] F. Duchˆ ene, C. Garbay, and V. Rialle, “Similarity measure for heterogeneous multivariate time-series,” in 2004 12th European Signal Processing Conference, Sep. 2004, pp. 1605–1608.

[34] S. Atev, G. Miller, and N. P. Papanikolopoulos, “Clustering of Vehicle Trajectories,” IEEE Transactions on Intelligent Transportation Systems, vol. 11, no. 3, pp. 647–657, Sep. 2010.

[35] S. L. Seyler, A. Kumar, M. F. Thorpe, and O. Beckstein, “Path Similarity Analysis: A Method for Quantifying Macromolecular Pathways,” PLOS Computational Biology, vol. 11, no. 10, p. e1004568, Oct. 2015, publisher: Public Library of Science. [Online]. Available: https://journals.plos.org/ploscompbiol/article?id= 10.1371/journal.pcbi.1004568

[36] J. Weise and S. Mostaghim, “A comparison of distance metrics for the multi-objective pathfinding problem,” Natural Computing, vol. 22, no. 2, pp. 315–328, Jun. 2023. [Online]. Available:

https://doi.org/10.1007/s11047-022-09908-z

[37] J. Kerber, S. Wagner, K. Groh, D. Notz, T. K¨ uhbeck, D. Watzenig, and A. Knoll, “Clustering of the Scenario Space for the Assessment of Automated Driving,” in 2020 IEEE Intelligent Vehicles Symposium (IV), Oct. 2020, pp. 578–583, iSSN: 2642-7214.

[38] J. Bernhard, M. Schutera, and E. Sax, “Optimizing test-set diversity: Trajectory clustering for scenario-based testing of automated driving systems,” in 2021 IEEE International Intelligent Transportation Systems Conference (ITSC), Sep. 2021, pp. 1371–1378.

[39] P. Nie, Z. Chen, N. Xia, Q. Huang, and F. Li, “Trajectory Similarity Analysis with the Weight of Direction and k-Neighborhood for AIS Data,” ISPRS International Journal of Geo-Information, vol. 10, no. 11, p. 757, Nov. 2021, number: 11 Publisher: Multidisciplinary Digital Publishing Institute. [Online]. Available: https://www.mdpi.com/2220-9964/10/11/757

[40] A.S. Jonkman, “Geneartion of Novel Critical Scenarios for Safety Assessment of Automated Driving Systems,” Eindhoven University of Technology, Eindhoven, Master’s Thesis, Aug. 2020.

[41] “Metric space,” page Version ID: 1165104782. [Online]. Available:

https://en.wikipedia.org/w/index.php?title=Metric space& oldid=1165104782

[42] M. M. Suarez-Alvarez, D.-T. Pham, M. Y. Prostov, and Y. I. Prostov, "Statistical approach to normalization of feature vectors and clustering of mixed datasets," Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, vol. 468, no. 2145, pp. 2630– 2651, Apr. 2012, publisher: Royal Society. [Online]. Available: https://royalsocietypublishing.org/doi/full/10.1098/rspa.2011.0704

[43] "Continuous or discrete variable," Jun. 2023, page Version ID: 1162238256. [Online]. Available: https://en.wikipedia.org/w/index. php?title=Continuous or discrete variable&oldid=1162238256

[44] "Cosine similarity," page Version ID: 1163816200. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Cosine similarity& oldid=1163816200

[45] Peter J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," Journal of Computational and Applied Mathematics, vol. 20, pp. 53–65, Nov. 1987.

[Online]. Available: https://www.sciencedirect.com/science/article/pii/0377042787901257